TIE TIY — Phase 3 Audits & vp Leakage Fix

Consultant Briefing | Round 9 | Audit body of evidence + Bear-only deployment question

Executive Summary

Per consultant Round 8 guidance, the team executed five harness audits on rule_019 (Bear UP_TRI hot refinement, the system's strongest validated edge candidate), discovered and surgically fixed a real look-ahead leakage in the volatility percentile feature, then re-validated the full 37-rule library on point-in-time-correct data. This document summarizes findings and surfaces one open strategic question for guidance.

Headline outcomes:

- Five harness audits completed on rule_019. Body of evidence is favorable: edge survives all perturbations, with one AMBIGUOUS verdict that has a benign explanation.
- Real bug found and fixed: **vp leakage** in feature_extractor._nifty_vol_percentile_at(). Forward-looking reference distribution corrected to point-in-time. Empirical impact on rule_019 is contained: lift dropped from +17.22pp to +14.77pp, classification unchanged (still SURVIVOR).
- Full 37-rule library re-validated post-fix. **Zero net change** in deployable rule count. No new SURVIVOR rules emerged for Bull or Choppy regimes despite ~10,360 sub_regime reclassifications across the dataset.
- **Open strategic question:** the deployable rule library remains Bear-regime-only (rule_019 + rule_031, plus kill_001 filter). vp fix did not unlock multi-regime coverage. Path forward unclear.

Document scope: sections 1-3 detail Phase 3 audit findings; section 4 covers vp leakage discovery and fix; section 5 covers post-fix re-validation results; section 6 lists the open consultation question.

1. Phase 3 Audit Methodology

Phase 3 was scoped per consultant Round 8 to test rule_019's edge against five perturbation classes that would expose specific leakage or contamination modes. Each audit is a separate Python script in lab/factory/oos_validation/, version-controlled, with deterministic numeric outputs persisted to JSON for reproducibility.

Rule under audit:

rule_019_bear_uptri_hot_refinement — fires when regime=Bear, signal=UP_TRI, sub_regime=hot. Pre-fix walk-forward: +17.22pp mean lift, 78% positive windows over 23 match-windows in 182 60-day rolling windows.

Reference distribution:

All audits cross-checked against a single shuffle-null reference: 100 global won-shuffles produced shuffle_p95 = 0.0503 (decimal), shuffle_max = 0.0867. Edge is statistically meaningful when lift > shuffle_p95.

2. Phase 3 Audit Results — Body of Evidence

Audit	Verdict	Key Number	Interpretation
3.2 label_randomization	PASS	z=5.01	Edge survives outcome shuffling. Real lift (17.22pp) >> shuffle_max (8.67pp).
3.3 date_shift Trial +1	PASS	drop=75.5%	Forward shift collapses edge as expected.
3.3 date_shift Trial -1	AMBIGUOUS	drop=16%	Backward shift preserves 84% of edge. See Section 3 for benign explanation.
3.3.7 vp leakage diag	BUG FOUND	126 trades	Real forward-looking leakage in vp feature. Empirically inert for rule_019 edge driver
3.4 universe_perturbation	PASS	162+ symbols	Edge broad-based. Lift INCREASES at top-N=1,3,5; barely declines at N=20.
3.5 corp_action_spike	PASS	0.66% trades	15 hard-threshold trades = COVID/crisis-recovery rallies, not corp actions.
3.6 embargo_sensitivity	AMBIG. / FAVORABLE	drift +6pp	Edge IMPROVES under reduced window overlap. Opposite of contamination signatu

Summary: 4 PASS, 2 AMBIGUOUS (both with substantive favorable interpretations), 1 real bug found (separately quantified as inert for rule_019). No audit produced a smoking-gun result that would invalidate rule_019's edge.

3. Key Audit Details

3.1 Label randomization (PASS)

100 global won-column shuffle trials, full walk-forward on shuffled outcomes. Real rule_019 lift = 17.22pp. Shuffle distribution: mean=-0.07pp, std=2.86pp, p95=5.03pp, max=8.67pp. z-score = 5.01. Real lift is 5+ standard deviations above the noise band. Edge is not an artifact of outcome distribution.

3.2 Date shift (mixed verdict)

Shifted scan_date by ±1 trading day. **Trial +1 (forward):** drop_pct = 75.5%, lift collapsed to 4.21pp (z=1.43). PASS — temporal alignment is required for the edge. **Trial -1 (backward):** drop_pct only 16%, lift = 14.41pp (z=4.25). AMBIGUOUS by spec.

Diagnostic chain for Trial -1: ran follow-up audits to identify the explanation. Hypothesis A (cell-specific autocorrelation) was REFUTED — rule_019 pooled autocorr 0.4407 vs broad cell baseline 0.4208 (delta only +0.020). Hypothesis B (vp leakage as driver) was REFUTED — leakage subset is 126 trades (5.5%) at 67.29% WR vs genuine 2,149 trades at 68.32% WR; removing the leakage moves WR by only +0.05pp.

Most likely explanation: Hypothesis E — broad within-symbol autocorrelation (~0.37 pooled across all signals system-wide). When backward-shifting outcomes, the next signal's outcome is correlated with the current's because outcomes cluster within symbols. Real momentum/persistence in Indian equity outcomes broadly, not specific to rule_019.

3.3 Universe perturbation (PASS)

Sequentially dropped top-N symbols by sum of pnl_pct contribution (N ∊ {1, 3, 5, 10, 20}), re-ran walk-forward. Edge trajectory:

N dropped	Matches dropped	Lift	Ratio vs real	Above shuffle p95
1	17	17.28pp	1.003	Yes (3.4×)
3	45	17.30pp	1.005	Yes
5	66	17.26pp	1.002	Yes
10	154	16.68pp	0.969	Yes
20	294	16.19pp	0.940	Yes (3.2×)

Strongest robustness signal observed: at N=1, 3, 5 lift INCREASES slightly — meaning top-pnl symbols had win rates roughly equal to the rule's average. Their dominance came from more matches and/or larger per-trade returns, not outsized win rates. Edge generalizes across 162+ distinct symbols. Top-pnl cluster: pharma sector (LUPIN, ZYDUSLIFE, GLENMARK, CIPLA, AUROPHARMA, LAURUSLABS).

3.4 Corporate action spike (PASS)

Identified rule_019 trades with anomalously large pnl. enriched_signals lacks per-day return data, so threshold operates on aggregate pnl_pct (entry-to-exit). Hard threshold pnl > 30%, soft 20-30%, negative < -20%.

Category	n	% of total	Within-subset WR
Hard threshold (>30%)	15	0.66%	100%
Soft threshold (20-30%)	42	1.85%	100%
Negative spike (<-20%)	2	0.09%	0%

WR-based edge impact: baseline WR 0.6827 → clean WR 0.6804 → wr_change_pp = 0.23pp (well below 1.0pp PASS limit). **Lift-based:** 17.22pp → 17.19pp (Δ 0.03pp).

Substantive finding (not corp-action contamination): the 15 hard-threshold trades are V-shaped recovery rallies at major macro selloff bottoms — 13 of 15 in COVID March-April 2020 (pharma vaccine optimism), 1 in 2013-08-21 (rupee crisis bottom), 1 in 2018-10-26 (NBFC crisis aftermath). exit_day = 6 with pnl_per_day 5-11% indicates sustained multi-day rallies, not single-day price spikes from splits/dividends. Negative spikes (n=2) are both YESBANK during RBI restructuring crash. The rule is working as designed: catching reversals from distressed conditions is its character.

3.5 Embargo sensitivity (AMBIGUOUS by spec, FAVORABLE empirically)

Naming caveat: consultant's classic train/test embargo concept doesn't apply directly — walkforward.py has no train/test split (each 60-day window evaluates lift = match_wr - same-window-baseline_wr self-contained). Audit reframed as *window-overlap sensitivity*: subsampled baseline windows[::N] to widen effective stride, testing whether 50% overlap of consecutive 60d/30d-step windows inflates aggregate stability metrics via autocorrelation.

Step (days)	n_windows	n_match_win	Lift	Signed drift	Decision
30 (baseline)	182	23	17.22pp	0	PASS
60	91	10	19.24pp	+2.02pp	PASS
90	61	6	23.37pp	+6.15pp	AMBIGUOUS
120	46	6	19.81pp	+2.59pp	PASS
150 (low-win)	37	3	12.68pp	-4.54pp	AMBIGUOUS
180 (low-win)	31	2	11.39pp	-5.83pp	AMBIGUOUS

Empirical reading: reducing window overlap INCREASES measured lift in the core trials (step 60, 90, 120), opposite of the contamination signature this test was designed to catch. If 50% overlap had been inflating recorded metrics through autocorrelation, reducing overlap should have collapsed lift toward shuffle_p95. Instead, lift rises. High-stride trials (150, 180) decline but with only 2-3 matched windows — not statistically meaningful, low-window warnings fired as expected. All 6 trials remain above shuffle_p95.

Conclusion across all five audits: rule_019's edge is empirically robust to the perturbations tested. One real bug found (vp leakage), separately quantified as inert for rule_019's edge driver. No smoking gun.

4. vp Leakage — Discovery, Fix, Impact

4.1 The bug

Discovered during Phase 3.3 sub_regime computation audit. The function **_nifty_vol_percentile_at(scan_date)** in lab/infrastructure/feature_extractor.py computed Nifty 20-day vol's percentile rank against the FULL 15-year vol distribution — including dates after scan_date. At scan_date 2014-01-15, the rank was computed against vol values from 2020+ that hadn't happened yet. Forward-looking by construction.

Pre-fix code (buggy):

```
log_ret = np.log(window / window.shift(1))
current = log_ret.iloc[-20:].std() * sqrt(252)
full_log_ret = np.log(self._nifty_close / self._nifty_close.shift(1)) # full history!
full_vol = full_log_ret.rolling(20).std() * sqrt(252) # full history!
rank = (full_vol <= current).sum() / len(full_vol)
```

Post-fix code (DRY): uses already-truncated log_ret as reference distribution.

```
log_ret = np.log(window / window.shift(1)) # window already truncated to <= scan_date
current = log_ret.iloc[-20:].std() * sqrt(252)
full_vol = log_ret.rolling(20).std() * sqrt(252) # PiT-correct
rank = (full_vol <= current).sum() / len(full_vol)
```

4.2 Verification of fix

Computed corrected vp on 4 verification dates, compared to point-in-time reference values from Phase 3.3.7 leakage_quantification.json. All matched within floating-point tolerance:

Scan date	Pre-fix (leaky)	Post-fix (PiT)	Diff	Within tol
2014-01-15	0.4580	0.1567	-0.3012	Yes
2018-01-15	0.0102	0.0082	-0.0019	Yes
2021-01-15	0.6930	0.6631	-0.0299	Yes
2024-01-15	0.2532	0.2271	-0.0261	Yes

Leakage magnitude correlates with date — early dates (2014) show 0.30+ gap because the post-2014 vol distribution is rich in COVID-2020 high-vol observations not yet realized. Recent dates (2024) show small gaps. Pattern matches Phase 3.3.7's empirical quantification.

4.3 Dataset-wide impact

Recomputed vp for all 105,987 signals using corrected function. Diff statistics:

vp absolute change	n signals	% of total
< 0.01	15,199	14.34%
0.01 to < 0.05	42,596	40.19%
0.05 to < 0.10	17,106	16.14%
≥ 0.10	31,086	29.33%

Mean absolute diff: 0.10. Max: 0.74. Nearly 30% of signals see vp shift by ≥0.10. Most large shifts are early-period signals (2011-2017).

Sub_regime transitions (10,360 rows = 9.77% changed):

From → To	Count	Notes

equilibrium_mid → quiet_mid	3,077	Choppy: vol over-rated
equilibrium_high → quiet_high	2,525	Choppy
stress_mid → equilibrium_mid	1,415	Choppy
warm → cold (Bear)	1,313	Bear: most affected non-rule_019 cell
stress_high → equilibrium_high	1,083	Choppy
hot → cold (Bear)	143	rule_019's affected cell
other transitions	~800	—

Pattern: almost all transitions are downgrades (higher-vol-tier → lower-vol-tier). Confirms Phase 3.3.7's finding that forward-looking vp over-rates vol percentile vs PiT. The 143 Bear hot→cold transitions match Phase 3.3.7's leakage_quantification of 126 leakage-inflated rule_019 trades within rounding (subset matched-and-resolved variant of total cell transitions).

4.4 Scope notes

Bug #2 deferred: a parallel forward-looking pattern exists in _compute_nifty_vol_thresholds() which produces the categorical feat_nifty_vol_regime feature. Traced to NOT be in walkforward path (uses _validate_paths.py:choppy_subregime which reads numerical vp directly, not categorical regime). Documented for separate workstream.

6 dead Choppy rules flagged: sub_regime_constraint values like 'breadth_high' don't match any sub_regime emitted by _validate_paths.py:choppy_subregime. These rules fire on zero windows by construction. Pre-existing schema mismatch, separate from vp leakage. Out of scope.

5. Post-Fix Re-Validation Results

After the surgical fix and feature recomputation, re-ran walk-forward on the full 37-rule library and re-ran W3 stability classification. Pre-fix outputs preserved as backups.

5.1 Category counts

Category	Pre-fix	Post-fix	Δ
SURVIVOR	2	2	0
WEAK_SURVIVOR	2	2	0
SURVIVOR_KILL	2	3	+1
WEAK_SURVIVOR_KILL	4	3	-1
DEGRADER + REJECT (others)	27	27	0
TOTAL	37	37	0

Net deployable change: 0 rules. Only one classification flipped: rule_013 promoted from WEAK_SURVIVOR_KILL to SURVIVOR_KILL (kill rule strengthened post-fix).

5.2 rule_019 specific (the rule of audit interest)

State	Category	Lift	Pos%	n_match_windows
Pre-fix	SURVIVOR	+17.22pp	78%	23
Post-fix	SURVIVOR (unchanged)	+14.77pp	77%	22
Δ	(unchanged)	-2.45pp	-1pp	-1

Lift dropped 2.45pp. Larger than Phase 3.3.7's WR-based prediction (~0.05pp WR shift) because the per-window lift mechanic differs from overall WR — the 126 leakage-inflated trades cluster in specific high-lift windows; removing them reduces those windows' match_wr more than the global average. Still comfortably SURVIVOR.

5.3 Top lift changes across rule library

Rule	Pre-cat	Post-cat	Δ lift
rule_028 bear_uptri_metal_hot	WEAK_SURVIVOR	WEAK_SURVIVOR	+3.97pp
rule_033 bear_bullproxy_warm	WATCH_INFO	WATCH_INFO	+3.73pp
rule_029 bear_uptri_pharma_hot	DEGRADER	DEGRADER	-3.72pp
rule_036 bear_uptri_bank_cold	REJECT_KILL_NOT_KILLING	(same)	+3.25pp
rule_019 bear_uptri_hot	SURVIVOR	SURVIVOR	-2.45pp
rule_013 bear_downtri_kill	WEAK_SURVIVOR_KILL	SURVIVOR_KILL	-1.77pp
rule_022 bear_uptri_warm	WATCH_INFO	WATCH_INFO	+0.97pp
rule_031 bear_uptri_it_hot	SURVIVOR	SURVIVOR	0.00pp
28 other rules	—	—	0.00pp

Sister Bear-hot rules diverged: rule_019 (no sector) -2.45pp, rule_031 (IT sector) 0.00pp, rule_028 (Metal) +3.97pp, rule_029 (Pharma) -3.72pp. Direction correlates with which symbols' rule_019 hot→cold transitions cluster — pharma + general had positive-pnl trades reclassified out, metal had marginal trades reclassified out.

5.4 Production-deployment intersection

wf × backtest intersection unchanged at 3 rules: kill_001 (Bank+DOWN_TRI filter, n=11 WR=0), win_001 (kept rule), rule_019 (Bear UP_TRI hot). All READY_TO_SHIP per W3 classification.

Per-regime deployable rule inventory (post-fix):
Bear: rule_019 (+14.77pp, 77%), rule_031 (+16.10pp, 79%) — 2 rules
Bull: 0 rules
Choppy: 0 positive-edge rules (rule_027 is kill only)
Filters: kill_001 (Bank+DOWN_TRI)

6. Open Strategic Question

6.1 Context

Trading has been paused since April 2026 pending validation. Phase 3 audits and vp leakage fix are now complete. The team's expectation entering the vp fix workstream was that correcting the look-ahead leakage might unlock new SURVIVOR rules in Bull and Choppy regimes (since the leakage was systemic, not rule_019-specific). That hypothesis did not hold: zero net change in deployable rule count.

6.2 The Bear-only constraint

rule_019 fires only when regime=Bear AND signal=UP_TRI AND sub_regime=hot. Bear regime is rare in the historical record (most years are Bull or Choppy). Hot sub_regime (nifty_60d_return_pct < -0.10 AND nifty_vol_percentile_20d > 0.70) is rarer still. Over 15 years, rule_019 fires only in 22 of 182 60-day windows (12.1% of historical periods). Even rule_031 (Bear UP_TRI IT-sector hot) is similarly rare-firing.

Practically: if markets are in Bull or Choppy regime, the deployed system produces ZERO signals. Months of idle periods are expected.

6.3 The pre-existing Round 8 framing

Consultant Round 8 anticipated this Bear-inactive scenario and recommended 'idle + parallel research' during inactive periods. The team's concern: 'idle' in production translates to 'no trading income.' For a sole operator who has paused trading since April, extended idle periods are non-trivially costly even if they're not loss-making.

6.4 Three paths under consideration

Path	Description	Estimated horizon
(A) Deploy Bear-only Phase 1	Deploy rule_019 + rule_031 + kill_001 alone with explicit acknowledgment of Bear-only coverage. 30-day shadow mo	4-6 weeks to live
(B) Re-run rule discovery	Re-run the 5+ weeks of Lab work (rule generation pipeline) with corrected vp + sub_regime data. May produce Bull/C	8-12 weeks
(C) New scanner architecture	Operator has a Swing Scanner blueprint (separate document) with cleaner separation of concerns and explicit point-i	14-18 weeks

6.5 Operator's specific question

What is the consultant's recommended path forward, given:

1. Phase 3 audit body of evidence (favorable for rule_019 specifically),
2. The vp leakage fix outcome (real bug fixed, but no multi-regime unlock),
3. The Bear-only deployment constraint (rule library covers only one regime),
4. Operator's runway / restart-trading priority (paused since April).

Specific sub-questions that would help shape the decision:

(i) Is Bear-only Phase 1 deployment a defensible first step, or is multi-regime coverage a pre-deployment requirement?

(ii) If re-running rule discovery with corrected data is the recommended path, what's the framework for assessing whether discovery itself was contaminated by the original leaky vp (and thus whether even a clean re-run can be trusted without further audit)?

(iii) What additional audit or evidence would strengthen confidence enough to deploy rule_019 + rule_031 (Bear-only) into shadow mode now while parallel work continues?

(iv) Is rule_031 (Bear UP_TRI IT-sector hot) deployable without its own Phase 3 audit cycle, or does it require separate validation before joining rule_019 in deployment?

All Phase 3 audit scripts, JSON outputs, vp fix code diff, pre-fix backups, and walk-forward outputs are version-controlled on the vp-leakage-fix branch and available for review on request.